



Equations from Quantities Teaching Bundle

Grade 7 • North Carolina • Mathematics

NC.7.EE.4 • Model and solve multi-step equations.

A focused classroom collection: task cards, five warm-ups, classwork, an exit ticket, four seasonal practice sets, substitute lessons, and planning pages. Each section has worked answers.

PRINT • TEACH • REVISIT



Your teaching route

Focus

Model and solve multi-step equations. This bundle practices a focused part of NC.7.EE.4; it is not a complete grade curriculum or a simulated EOG.

Before teaching

Learners should explain a simpler example using a model. Read the two worked examples, then use the first task card to check readiness.

Suggested sequence

Day 1: model for 10 minutes, then guided task cards for 15 minutes. Day 2: a 5-minute warm-up and 20 minutes of practice. Use the exit ticket to choose a small-group reteaching focus. Seasonal sets revisit the same skill using different quantities.

Materials

Pencils, blank paper, and graph paper or counters as useful. No paid app or internet is needed for student work. A calculator may check finished calculations; these are instructional choices, not official testing accommodation rules.



Make the reasoning visible

Define the unknown with units. Undo addition before multiplication, or expand first. Substitute the answer into the original equation.

Example 1: The school garden pays a \$2 setup fee plus \$4 for each kit. The total is \$78. Write and solve an equation for the number of kits.

Let x count kits. $4x + 2 = 78$; $4x = 76$; $x = 19$. Check: $4(19) + 2 = 78$.

Example 2: The reading group pays a \$5 setup fee plus \$2 for each kit. The total is \$27. Write and solve an equation for the number of kits.

Let x count kits. $2x + 5 = 27$; $2x = 22$; $x = 11$. Check: $2(11) + 5 = 27$.

Ask: What does each number represent? How could a model or inverse operation check the result?

Task cards 1–4

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 01

The reading group pays a \$6 setup fee plus \$6 for each kit. The total is \$84. Write and solve an equation for the number of kits.

Explain your method.

CARD 02

The library team pays a \$3 setup fee plus \$6 for each kit. The total is \$39. Write and solve an equation for the number of kits.

Explain your method.

CARD 03

The design team pays a \$8 setup fee plus \$4 for each kit. The total is \$84. Write and solve an equation for the number of kits.

Explain your method.

CARD 04

The community center pays a \$5 setup fee plus \$8 for each kit. The total is \$125. Write and solve an equation for the number of kits.

Explain your method.



Task cards 5–8

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 05

The school garden pays a \$2 setup fee plus \$5 for each kit. The total is \$52. Write and solve an equation for the number of kits.

Explain your method.

CARD 06

The art club pays a \$7 setup fee plus \$9 for each kit. The total is \$88. Write and solve an equation for the number of kits.

Explain your method.

CARD 07

The design team pays a \$8 setup fee plus \$3 for each kit. The total is \$47. Write and solve an equation for the number of kits.

Explain your method.

CARD 08

The community center pays a \$3 setup fee plus \$5 for each kit. The total is \$48. Write and solve an equation for the number of kits.

Explain your method.



Task cards 9–12

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 09

The library team pays a \$5 setup fee plus \$4 for each kit. The total is \$73. Write and solve an equation for the number of kits.

Explain your method.

CARD 10

The science club pays a \$6 setup fee plus \$3 for each kit. The total is \$33. Write and solve an equation for the number of kits.

Explain your method.

CARD 11

The design team pays a \$5 setup fee plus \$2 for each kit. The total is \$33. Write and solve an equation for the number of kits.

Explain your method.

CARD 12

The library team pays a \$3 setup fee plus \$8 for each kit. The total is \$115. Write and solve an equation for the number of kits.

Explain your method.



Task cards 13–16

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 13

The school garden pays a \$5 setup fee plus \$9 for each kit. The total is \$104. Write and solve an equation for the number of kits.

Explain your method.

CARD 14

The reading group pays a \$6 setup fee plus \$5 for each kit. The total is \$91. Write and solve an equation for the number of kits.

Explain your method.

CARD 15

The library team pays a \$6 setup fee plus \$5 for each kit. The total is \$66. Write and solve an equation for the number of kits.

Explain your method.

CARD 16

The design team pays a \$3 setup fee plus \$8 for each kit. The total is \$75. Write and solve an equation for the number of kits.

Explain your method.

Task-card response record

Use this page for final answers; attach drawings or full worked solutions on separate paper.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____
11. _____
12. _____
13. _____
14. _____
15. _____
16. _____



Warm-up • Day 1

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

1. The reading group pays a \$2 setup fee plus \$9 for each kit. The total is \$29. Write and solve an equation for the number of kits.

2. The reading group pays a \$5 setup fee plus \$3 for each kit. The total is \$32. Write and solve an equation for the number of kits.



Warm-up • Day 2

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

3. The music club pays a \$6 setup fee plus \$6 for each kit. The total is \$48. Write and solve an equation for the number of kits.

4. The reading group pays a \$7 setup fee plus \$8 for each kit. The total is \$159. Write and solve an equation for the number of kits.



Warm-up • Day 3

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

5. The community center pays a \$3 setup fee plus \$3 for each kit. The total is \$42. Write and solve an equation for the number of kits.

6. The reading group pays a \$7 setup fee plus \$3 for each kit. The total is \$55. Write and solve an equation for the number of kits.



Warm-up • Day 4

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

7. The music club pays a \$2 setup fee plus \$7 for each kit. The total is \$121. Write and solve an equation for the number of kits.

8. The community center pays a \$2 setup fee plus \$7 for each kit. The total is \$121. Write and solve an equation for the number of kits.



Warm-up • Day 5

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

9. The library team pays a \$8 setup fee plus \$8 for each kit. The total is \$32. Write and solve an equation for the number of kits.

10. The reading group pays a \$8 setup fee plus \$6 for each kit. The total is \$62. Write and solve an equation for the number of kits.



Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. The library team pays a \$6 setup fee plus \$4 for each kit. The total is \$50. Write and solve an equation for the number of kits.

2. The music club pays a \$7 setup fee plus \$3 for each kit. The total is \$37. Write and solve an equation for the number of kits.

3. The science club pays a \$5 setup fee plus \$2 for each kit. The total is \$41. Write and solve an equation for the number of kits.



Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. The design team pays a \$2 setup fee plus \$5 for each kit. The total is \$17. Write and solve an equation for the number of kits.

5. The design team pays a \$2 setup fee plus \$2 for each kit. The total is \$14. Write and solve an equation for the number of kits.

6. The art club pays a \$3 setup fee plus \$8 for each kit. The total is \$35. Write and solve an equation for the number of kits.



Exit ticket • Try independently

Complete the four questions independently. Show enough reasoning for your teacher to see your method. This is a classroom check, not a standardized score.

1. The science club pays a \$2 setup fee plus \$4 for each kit. The total is \$14. Write and solve an equation for the number of kits.

2. The library team pays a \$6 setup fee plus \$3 for each kit. The total is \$27. Write and solve an equation for the number of kits.

3. The community center pays a \$6 setup fee plus \$3 for each kit. The total is \$15. Write and solve an equation for the number of kits.

4. The community center pays a \$8 setup fee plus \$4 for each kit. The total is \$56. Write and solve an equation for the number of kits.



Task-card worked solutions

1. 13 kits

Let x count kits. $6x + 6 = 84$; $6x = 78$; $x = 13$. Check: $6(13) + 6 = 84$.

2. 6 kits

Let x count kits. $6x + 3 = 39$; $6x = 36$; $x = 6$. Check: $6(6) + 3 = 39$.

3. 19 kits

Let x count kits. $4x + 8 = 84$; $4x = 76$; $x = 19$. Check: $4(19) + 8 = 84$.

4. 15 kits

Let x count kits. $8x + 5 = 125$; $8x = 120$; $x = 15$. Check: $8(15) + 5 = 125$.

5. 10 kits

Let x count kits. $5x + 2 = 52$; $5x = 50$; $x = 10$. Check: $5(10) + 2 = 52$.

6. 9 kits

Let x count kits. $9x + 7 = 88$; $9x = 81$; $x = 9$. Check: $9(9) + 7 = 88$.

Task-card worked solutions

7. 13 kits

Let x count kits. $3x + 8 = 47$; $3x = 39$; $x = 13$. Check: $3(13) + 8 = 47$.

8. 9 kits

Let x count kits. $5x + 3 = 48$; $5x = 45$; $x = 9$. Check: $5(9) + 3 = 48$.

9. 17 kits

Let x count kits. $4x + 5 = 73$; $4x = 68$; $x = 17$. Check: $4(17) + 5 = 73$.

10. 9 kits

Let x count kits. $3x + 6 = 33$; $3x = 27$; $x = 9$. Check: $3(9) + 6 = 33$.

11. 14 kits

Let x count kits. $2x + 5 = 33$; $2x = 28$; $x = 14$. Check: $2(14) + 5 = 33$.

12. 14 kits

Let x count kits. $8x + 3 = 115$; $8x = 112$; $x = 14$. Check: $8(14) + 3 = 115$.

Task-card worked solutions

13. 11 kits

Let x count kits. $9x + 5 = 104$; $9x = 99$; $x = 11$. Check: $9(11) + 5 = 104$.

14. 17 kits

Let x count kits. $5x + 6 = 91$; $5x = 85$; $x = 17$. Check: $5(17) + 6 = 91$.

15. 12 kits

Let x count kits. $5x + 6 = 66$; $5x = 60$; $x = 12$. Check: $5(12) + 6 = 66$.

16. 9 kits

Let x count kits. $8x + 3 = 75$; $8x = 72$; $x = 9$. Check: $8(9) + 3 = 75$.

Warm-up worked solutions

1. 3 kits

Let x count kits. $9x + 2 = 29$; $9x = 27$; $x = 3$. Check: $9(3) + 2 = 29$.

2. 9 kits

Let x count kits. $3x + 5 = 32$; $3x = 27$; $x = 9$. Check: $3(9) + 5 = 32$.

3. 7 kits

Let x count kits. $6x + 6 = 48$; $6x = 42$; $x = 7$. Check: $6(7) + 6 = 48$.

4. 19 kits

Let x count kits. $8x + 7 = 159$; $8x = 152$; $x = 19$. Check: $8(19) + 7 = 159$.

5. 13 kits

Let x count kits. $3x + 3 = 42$; $3x = 39$; $x = 13$. Check: $3(13) + 3 = 42$.

6. 16 kits

Let x count kits. $3x + 7 = 55$; $3x = 48$; $x = 16$. Check: $3(16) + 7 = 55$.

Warm-up worked solutions

7. 17 kits

Let x count kits. $7x + 2 = 121$; $7x = 119$; $x = 17$. Check: $7(17) + 2 = 121$.

8. 17 kits

Let x count kits. $7x + 2 = 121$; $7x = 119$; $x = 17$. Check: $7(17) + 2 = 121$.

9. 3 kits

Let x count kits. $8x + 8 = 32$; $8x = 24$; $x = 3$. Check: $8(3) + 8 = 32$.

10. 9 kits

Let x count kits. $6x + 8 = 62$; $6x = 54$; $x = 9$. Check: $6(9) + 8 = 62$.

Classwork worked solutions

1. 11 kits

Let x count kits. $4x + 6 = 50$; $4x = 44$; $x = 11$. Check: $4(11) + 6 = 50$.

2. 10 kits

Let x count kits. $3x + 7 = 37$; $3x = 30$; $x = 10$. Check: $3(10) + 7 = 37$.

3. 18 kits

Let x count kits. $2x + 5 = 41$; $2x = 36$; $x = 18$. Check: $2(18) + 5 = 41$.

4. 3 kits

Let x count kits. $5x + 2 = 17$; $5x = 15$; $x = 3$. Check: $5(3) + 2 = 17$.

5. 6 kits

Let x count kits. $2x + 2 = 14$; $2x = 12$; $x = 6$. Check: $2(6) + 2 = 14$.

6. 4 kits

Let x count kits. $8x + 3 = 35$; $8x = 32$; $x = 4$. Check: $8(4) + 3 = 35$.

Exit-ticket worked solutions

1. 3 kits

Let x count kits. $4x + 2 = 14$; $4x = 12$; $x = 3$. Check: $4(3) + 2 = 14$.

2. 7 kits

Let x count kits. $3x + 6 = 27$; $3x = 21$; $x = 7$. Check: $3(7) + 6 = 27$.

3. 3 kits

Let x count kits. $3x + 6 = 15$; $3x = 9$; $x = 3$. Check: $3(3) + 6 = 15$.

4. 12 kits

Let x count kits. $4x + 8 = 56$; $4x = 48$; $x = 12$. Check: $4(12) + 8 = 56$.



Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. The fall garden club pays a \$2 setup fee plus \$8 for each kit. The total is \$114. Write and solve an equation for the number of kits.

2. The leaf observation team pays a \$7 setup fee plus \$6 for each kit. The total is \$55. Write and solve an equation for the number of kits.

3. The autumn reading fair pays a \$2 setup fee plus \$7 for each kit. The total is \$23. Write and solve an equation for the number of kits.

4. The October book drive pays a \$2 setup fee plus \$6 for each kit. The total is \$92. Write and solve an equation for the number of kits.



Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. The September art display pays a \$7 setup fee plus \$3 for each kit. The total is \$64. Write and solve an equation for the number of kits.

6. The leaf observation team pays a \$3 setup fee plus \$4 for each kit. The total is \$79. Write and solve an equation for the number of kits.

7. The autumn reading fair pays a \$8 setup fee plus \$5 for each kit. The total is \$53. Write and solve an equation for the number of kits.

8. The autumn reading fair pays a \$5 setup fee plus \$4 for each kit. The total is \$57. Write and solve an equation for the number of kits.



Fall • Worked solutions

1. 14 kits

Let x count kits. $8x + 2 = 114$; $8x = 112$; $x = 14$. Check: $8(14) + 2 = 114$.

2. 8 kits

Let x count kits. $6x + 7 = 55$; $6x = 48$; $x = 8$. Check: $6(8) + 7 = 55$.

3. 3 kits

Let x count kits. $7x + 2 = 23$; $7x = 21$; $x = 3$. Check: $7(3) + 2 = 23$.

4. 15 kits

Let x count kits. $6x + 2 = 92$; $6x = 90$; $x = 15$. Check: $6(15) + 2 = 92$.



Fall • Worked solutions

5. 19 kits

Let x count kits. $3x + 7 = 64$; $3x = 57$; $x = 19$. Check: $3(19) + 7 = 64$.

6. 19 kits

Let x count kits. $4x + 3 = 79$; $4x = 76$; $x = 19$. Check: $4(19) + 3 = 79$.

7. 9 kits

Let x count kits. $5x + 8 = 53$; $5x = 45$; $x = 9$. Check: $5(9) + 8 = 53$.

8. 13 kits

Let x count kits. $4x + 5 = 57$; $4x = 52$; $x = 13$. Check: $4(13) + 5 = 57$.



Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. The indoor garden project pays a \$7 setup fee plus \$3 for each kit. The total is \$46. Write and solve an equation for the number of kits.

2. The winter reading club pays a \$8 setup fee plus \$3 for each kit. The total is \$38. Write and solve an equation for the number of kits.

3. The December art display pays a \$4 setup fee plus \$9 for each kit. The total is \$157. Write and solve an equation for the number of kits.

4. The indoor garden project pays a \$8 setup fee plus \$8 for each kit. The total is \$48. Write and solve an equation for the number of kits.



Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. The winter reading club pays a \$4 setup fee plus \$2 for each kit. The total is \$18. Write and solve an equation for the number of kits.

6. The winter supply drive pays a \$2 setup fee plus \$4 for each kit. The total is \$66. Write and solve an equation for the number of kits.

7. The indoor garden project pays a \$2 setup fee plus \$7 for each kit. The total is \$44. Write and solve an equation for the number of kits.

8. The January puzzle club pays a \$8 setup fee plus \$4 for each kit. The total is \$72. Write and solve an equation for the number of kits.



Winter • Worked solutions

1. 13 kits

Let x count kits. $3x + 7 = 46$; $3x = 39$; $x = 13$. Check: $3(13) + 7 = 46$.

2. 10 kits

Let x count kits. $3x + 8 = 38$; $3x = 30$; $x = 10$. Check: $3(10) + 8 = 38$.

3. 17 kits

Let x count kits. $9x + 4 = 157$; $9x = 153$; $x = 17$. Check: $9(17) + 4 = 157$.

4. 5 kits

Let x count kits. $8x + 8 = 48$; $8x = 40$; $x = 5$. Check: $8(5) + 8 = 48$.



Winter • Worked solutions

5. 7 kits

Let x count kits. $2x + 4 = 18$; $2x = 14$; $x = 7$. Check: $2(7) + 4 = 18$.

6. 16 kits

Let x count kits. $4x + 2 = 66$; $4x = 64$; $x = 16$. Check: $4(16) + 2 = 66$.

7. 6 kits

Let x count kits. $7x + 2 = 44$; $7x = 42$; $x = 6$. Check: $7(6) + 2 = 44$.

8. 16 kits

Let x count kits. $4x + 8 = 72$; $4x = 64$; $x = 16$. Check: $4(16) + 8 = 72$.

Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. The April science team pays a \$3 setup fee plus \$2 for each kit. The total is \$27. Write and solve an equation for the number of kits.

2. The March art display pays a \$5 setup fee plus \$9 for each kit. The total is \$176. Write and solve an equation for the number of kits.

3. The April science team pays a \$5 setup fee plus \$8 for each kit. The total is \$133. Write and solve an equation for the number of kits.

4. The seedling club pays a \$7 setup fee plus \$8 for each kit. The total is \$87. Write and solve an equation for the number of kits.



Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. The spring reading fair pays a \$8 setup fee plus \$8 for each kit. The total is \$32. Write and solve an equation for the number of kits.

6. The seedling club pays a \$3 setup fee plus \$5 for each kit. The total is \$53. Write and solve an equation for the number of kits.

7. The March art display pays a \$4 setup fee plus \$9 for each kit. The total is \$166. Write and solve an equation for the number of kits.

8. The spring reading fair pays a \$3 setup fee plus \$6 for each kit. The total is \$69. Write and solve an equation for the number of kits.



Spring break • Worked solutions

1. 12 kits

Let x count kits. $2x + 3 = 27$; $2x = 24$; $x = 12$. Check: $2(12) + 3 = 27$.

2. 19 kits

Let x count kits. $9x + 5 = 176$; $9x = 171$; $x = 19$. Check: $9(19) + 5 = 176$.

3. 16 kits

Let x count kits. $8x + 5 = 133$; $8x = 128$; $x = 16$. Check: $8(16) + 5 = 133$.

4. 10 kits

Let x count kits. $8x + 7 = 87$; $8x = 80$; $x = 10$. Check: $8(10) + 7 = 87$.



Spring break • Worked solutions

5. 3 kits

Let x count kits. $8x + 8 = 32$; $8x = 24$; $x = 3$. Check: $8(3) + 8 = 32$.

6. 10 kits

Let x count kits. $5x + 3 = 53$; $5x = 50$; $x = 10$. Check: $5(10) + 3 = 53$.

7. 18 kits

Let x count kits. $9x + 4 = 166$; $9x = 162$; $x = 18$. Check: $9(18) + 4 = 166$.

8. 11 kits

Let x count kits. $6x + 3 = 69$; $6x = 66$; $x = 11$. Check: $6(11) + 3 = 69$.



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. The summer supply team pays a \$6 setup fee plus \$8 for each kit. The total is \$86. Write and solve an equation for the number of kits.

2. The July puzzle camp pays a \$4 setup fee plus \$9 for each kit. The total is \$49. Write and solve an equation for the number of kits.

3. The community garden pays a \$4 setup fee plus \$5 for each kit. The total is \$64. Write and solve an equation for the number of kits.

4. The summer supply team pays a \$7 setup fee plus \$3 for each kit. The total is \$61. Write and solve an equation for the number of kits.



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. The July puzzle camp pays a \$7 setup fee plus \$8 for each kit. The total is \$47. Write and solve an equation for the number of kits.

6. The summer supply team pays a \$4 setup fee plus \$6 for each kit. The total is \$106. Write and solve an equation for the number of kits.

7. The July puzzle camp pays a \$8 setup fee plus \$6 for each kit. The total is \$92. Write and solve an equation for the number of kits.

8. The June art workshop pays a \$8 setup fee plus \$3 for each kit. The total is \$62. Write and solve an equation for the number of kits.



Summer • Worked solutions

1. 10 kits

Let x count kits. $8x + 6 = 86$; $8x = 80$; $x = 10$. Check: $8(10) + 6 = 86$.

2. 5 kits

Let x count kits. $9x + 4 = 49$; $9x = 45$; $x = 5$. Check: $9(5) + 4 = 49$.

3. 12 kits

Let x count kits. $5x + 4 = 64$; $5x = 60$; $x = 12$. Check: $5(12) + 4 = 64$.

4. 18 kits

Let x count kits. $3x + 7 = 61$; $3x = 54$; $x = 18$. Check: $3(18) + 7 = 61$.



Summer • Worked solutions

5. 5 kits

Let x count kits. $8x + 7 = 47$; $8x = 40$; $x = 5$. Check: $8(5) + 7 = 47$.

6. 17 kits

Let x count kits. $6x + 4 = 106$; $6x = 102$; $x = 17$. Check: $6(17) + 4 = 106$.

7. 14 kits

Let x count kits. $6x + 8 = 92$; $6x = 84$; $x = 14$. Check: $6(14) + 8 = 92$.

8. 18 kits

Let x count kits. $3x + 8 = 62$; $3x = 54$; $x = 18$. Check: $3(18) + 8 = 62$.



Substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Model and solve multi-step equations. Let students explain a familiar strategy to a partner.

5-15 minutes: model

Use the worked-example page. Define the unknown with units. Undo addition before multiplication, or expand first. Substitute the answer into the original equation.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. The library helpers pays a \$2 setup fee plus \$7 for each kit. The total is \$79. Write and solve an equation for the number of kits.

2. The school art team pays a \$7 setup fee plus \$2 for each kit. The total is \$41. Write and solve an equation for the number of kits.

3. The school art team pays a \$4 setup fee plus \$2 for each kit. The total is \$32. Write and solve an equation for the number of kits.

4. The school art team pays a \$4 setup fee plus \$4 for each kit. The total is \$52. Write and solve an equation for the number of kits.



Substitute lesson • Worked solutions

1. 11 kits

Let x count kits. $7x + 2 = 79$; $7x = 77$; $x = 11$. Check: $7(11) + 2 = 79$.

2. 17 kits

Let x count kits. $2x + 7 = 41$; $2x = 34$; $x = 17$. Check: $2(17) + 7 = 41$.

3. 14 kits

Let x count kits. $2x + 4 = 32$; $2x = 28$; $x = 14$. Check: $2(14) + 4 = 32$.

4. 12 kits

Let x count kits. $4x + 4 = 52$; $4x = 48$; $x = 12$. Check: $4(12) + 4 = 52$.

Emergency substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Model and solve multi-step equations. Let students explain a familiar strategy to a partner.

5-15 minutes: model

No copies or devices? Write the first worked example on the board; students use blank paper. Define the unknown with units. Undo addition before multiplication, or expand first. Substitute the answer into the original equation.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Emergency substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. The paper puzzle team pays a \$5 setup fee plus \$3 for each kit. The total is \$38. Write and solve an equation for the number of kits.

2. The desk-supply team pays a \$8 setup fee plus \$8 for each kit. The total is \$40. Write and solve an equation for the number of kits.

3. The desk-supply team pays a \$2 setup fee plus \$8 for each kit. The total is \$162. Write and solve an equation for the number of kits.

4. The paper puzzle team pays a \$6 setup fee plus \$6 for each kit. The total is \$108. Write and solve an equation for the number of kits.



Emergency substitute lesson • Worked solutions

1. 11 kits

Let x count kits. $3x + 5 = 38$; $3x = 33$; $x = 11$. Check: $3(11) + 5 = 38$.

2. 4 kits

Let x count kits. $8x + 8 = 40$; $8x = 32$; $x = 4$. Check: $8(4) + 8 = 40$.

3. 20 kits

Let x count kits. $8x + 2 = 162$; $8x = 160$; $x = 20$. Check: $8(20) + 2 = 162$.

4. 17 kits

Let x count kits. $6x + 6 = 108$; $6x = 102$; $x = 17$. Check: $6(17) + 6 = 108$.

A completed lesson-plan example

Level and focus

Grade 7; NC.7.EE.4; Model and solve multi-step equations.

Evidence of learning

Student represents the quantities, selects a valid method, calculates accurately, and explains the result with appropriate units.

Launch / model

Use worked example 1. Ask students to annotate the role of every quantity. Demonstrate the model before moving to symbols.

Guided / independent practice

Use cards 1-4 in pairs; assign classwork independently. Change grouping according to the explanation you hear rather than speed alone.

Formative check

Use the four-question exit ticket. Score each item 0-2: 2 for an accurate result with coherent reasoning, 1 for a sound method with a minor error or an unexplained correct result, 0 for missing or unsupported reasoning.

Next teaching decision

0-3 points: reteach with a concrete model. 4-6: correct the specific misconception with a short example. 7-8: invite a second representation or a student-created problem. These are local instructional decisions, not mastery cutoffs validated by a test publisher.

Plan your next lesson

Date, class, and focus / exact local standard:

What will students do to show understanding?

Launch and worked example:

Guided practice and independent task:

Access supports and extension:

Exit evidence and next lesson decision:



Access, sources, and classroom license

AIG: Invite two methods, a counterexample, or a new task with a specified solution. Ask students to explain limits of a shortcut.

504: Apply the student's plan: breaks, extra time, enlarged print, reduced copying, or alternate response methods as documented.

EC: Use the IEP to select supports. Model one step, offer counters or a diagram, reduce the visible item set, and accept an oral explanation when appropriate.

ESL: Preview quantity, difference, factor, or other relevant terms. Let students rehearse in a home language, then use "I chose ___ because ___." Maintain the mathematical goal.

Official alignment sources • accessed September 15, 2026

NC.7.EE.4 is verified in the NCDPI mathematics progression. The activity objective is original wording; it describes the skill focus, not the full standard text.

NCDPI mathematics standards:

<https://www.dpi.nc.gov/documents/publications/catalog/ma196-vertical-progression-math/open>

NCDPI EOG mathematics specifications:

<https://www.dpi.nc.gov/documents/accountability/testing/eog/eog-mathematics-grades-3-8-test-specifications>

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